Deviations Register — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
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Version: 1.6 · **Date:** 2026-11-11 · **Covers:** Sprint 1 (19–23 Oct 2026), Sprint 2 (26 Oct – 1 Nov 2026, D-14), Deliverable 3 (2–3 Nov 2026) and final-demo preparation (11 Nov 2026)

1. Purpose

The Sprint Plan requires every departure from the SRS, SAD, Test Plan or Sprint Plan to be recorded here: what changed, why, and which requirement it affects. Each entry names the pull request or issue where the decision is visible in the repository history. Requirement and test-case IDs are the ones used in `docs/SRS.pdf` and `docs/Test_Plan.pdf`.

2. Summary

ID	Date	Baseline document	Affects	Kind	Status
D-01	22 Oct	SRS	FR-004, UT_014	Requirement interpretation	Accepted
D-02	22 Oct	SRS	FR-010, FR-023, NFR-010, UT_030	Design	Accepted
D-03	23 Oct	Sprint Plan	FR-019, FR-020	Scope moved earlier	Accepted
D-04	23–27 Oct	SRS (Organisational)	NFR-006	Process	Closed with exceptions; corrective action not kept in Sprint 2 (see D-08)
D-05	23–27 Oct	Sprint Plan	12 of 52 Sprint 1 points	Schedule	Resolved 27 Oct
D-06	23 Oct	Sprint Plan	Sprint 1 goal; FR-024, FR-027	Scope	Closed; delivered in Sprint 2
D-07	19–27 Oct	Sprint Plan (Definition of Done)	All Sprint 1 test cases	Process	Closed 3 Nov by Test Plan v4.0
D-08	27–30 Oct	SRS (Organisational)	NFR-006	Process	Closed with exceptions; repeats D-04
D-09	1 Nov	Sprint Plan	NFR-007, NFR-012	Reassignment	Accepted
D-10	1 Nov	SRS (Organisational)	NFR-006	Process	Logged under D-08's corrective action
D-11	3 Nov	SRS (Organisational)	NFR-006	Process	Logged under D-08's corrective action
D-12	3 Nov	Deliverable 3 work split	System Validation Report; NFR-006	Review correction; process	Accepted; admin merge logged
D-13	3 Nov	SRS, SAD, Test Plan, Sprint Plan	Document structure; SAD auto-submit; FR-028, ST_013; sprint videos	Documentation; defect carried forward; late deliverable	Accepted; Defect 1 closed by D-15
D-14	31 Oct – 1 Nov	Sprint Plan	23 of 45 Sprint 2 points; development freeze	Schedule	Accepted; freeze is build 2d74453
D-15	11 Nov	Frozen build; System Validation Report; Test Plan	FR-028, ST_013	Defect fix after the freeze	Closed; ST_013 Pass, 119 of 119

3. Deviations

D-01 — Course code is unique per semester, not globally

Baseline: FR-004 (SRS) says a duplicate course code is rejected.
Built: a course code is unique within a semester. The same code can be reused in a different semester. The database enforces this with the constraint `course_code_unique_per_semester`.
Why: the SRS and the Test Plan disagree. UT_014 expects the same code to be accepted in another semester, which matches how PES University reuses course codes every year. The team followed the test case.
Affects: FR-004, UT_014.
Recorded in: PR #71 (Komal), merged 22 Oct.

D-02 — Negative marking: toggle per examination, deduction per question

Baseline: FR-010 lists "negative marking (on/off and value)" as an examination policy value.
Built: the examination stores only the on/off switch (`Exam.negative_marking`). The deduction value is stored on each question (`Question.negative_marks`), next to its positive marks.
Why: FR-008 already gives every question its own marks. Keeping the deduction beside them lets a 1-mark and a 4-mark question carry different penalties in one paper. The evaluator applies a question's deduction only when the examination's switch is on, so a single paper can still have negative marking turned off. The SRS's open decision on negative marking (flat deduction of the question's configured negative marks) is unchanged.
Affects: FR-010, FR-023, NFR-010. UT_030 still passes: the switch and a 0.5 deduction persist and read back as set.
Recorded in: PR #74 (Komal), merged 22 Oct.

D-03 — Answer saving delivered one sprint early

Baseline: the Sprint Plan places FR-019 (autosave every response) in Sprint 2.
Built: PR #73 (FR-020, automatic submission) also added the service that saves an answer and its endpoint POST `/api/attempts/<id>/answers`. Each choice is saved the moment it is made.
Why: FR-020 submits the answers held at expiry. Without saved answers, an attempt closed by the timer would always score zero.
Affects: FR-019 (Sprint 2 now adds `marked_for_review` and the 5-second guarantee on top of the existing endpoint), FR-020.
Recorded in: PR #73 (Kshitij), merged 23 Oct.

D-04 — Five pull requests merged without a second member's approval

Baseline: NFR-006 requires every merge to `main` to go through a pull request reviewed by another member. Its acceptance test: no direct commits to `main` after 19 Oct, and every merged PR has a reviewer other than its author.
What happened: there were no direct commits to `main`. Nine of the fourteen pull requests merged since 19 Oct carry an **Approve** review from another member. Five of Kshitij's pull requests were merged with the repository owner's admin override instead:

PR	Item	Review	Merged
#68	FR-022, FR-023 evaluation	Komal commented ("Looks good"), did not approve	23 Oct
#70	FR-015, FR-016 start attempt	Komal commented ("Cool!"), did not approve	23 Oct
#73	FR-020 automatic submission	Komal commented ("Okay"), did not approve	23 Oct
#79	Start exam button on the student exam page	None	27 Oct
#80	Shared stylesheet and role navigation	None	27 Oct

Why: #68, #70 and #73 were stacked (each built on the one before) and blocked the Sprint 1 video due 23 Oct. #79 and #80 were merged before the Sprint 1 review demo on 27 Oct. CI was green on all five, and the full suite (109 tests) passes on `main`.

Corrective action: from Sprint 2 there are no admin merges. Every pull request, including the repository owner's, waits for an **Approve** review from another member; a comment does not count.
Recorded in: issue #38, closed 27 Oct with the full audit.

D-05 — Sprint 1 finished four days late

Baseline: Sprint 1 runs 19–23 Oct with a 52-point commitment.
What happened: 40 points closed by 23 Oct. The remaining 12 points were coded within the sprint but merged on 27 Oct:

Item	Points	Closed by	Cause
FR-006 enrolled-exam visibility	3	PR #75	Review asked for 403 instead of 404 (IT_011) and test IDs
FR-017 server-clock countdown	3	PR #76	Review asked it to reuse the attempt services; merge conflict
FR-029 no answer key in Student payloads	2	PR #76	Same pull request as FR-017
NFR-013 CSRF, ORM-only, generic error pages	2	PR #78	Waiting for review
NFR-014 secrets from environment variables	1	Verified, closed by hand	Waiting on #78
NFR-006 review by another member	1	Issue #38	See D-04

Merge conflicts were the main cause. A teammate's editor reformatted whole files on save (`repository/models.py` in PR #71, `attempts/views.py` in PR #75), so every branch touching those files conflicted. The review also found that the student exam page had no control to start an attempt (FR-006 and FR-015 were built by different members); PR #79 added it on 27 Oct.
Corrective action: members turn off format-on-save for this repository (requested in the review of PR #71), and a branch merges `main` into itself before review so conflicts are resolved by its owner.
Affects: the Sprint 1 burndown, which shows the drop on 27 Oct.

D-06 — The Sprint 1 score is computed but not yet shown to the Student

Baseline: the Sprint 1 goal ends with the Student "receiving an automatically computed score".
Built: every submission is evaluated automatically and its score is stored on the attempt (FR-022, FR-023). No screen shows it yet: the Student scorecard and the Faculty results view belong to FR-024 to FR-027, which the Sprint Plan places in Sprint 2. In the Sprint 1 demo the stored score is read from the database.
Affects: Sprint 1 goal; FR-024, FR-027.
Status: closed. Sprint 2 delivered both screens: FR-024's total, percentage and Pass / Fail (PR #83) and FR-027's Student scorecard (PR #100). Test cases UT_057, UT_058, UT_062 and UT_063 pass in Test Plan v4.0.

D-07 — Test results recorded in pull requests, not in the Test Plan

Baseline: Definition of Done item 3 requires each item's test cases in `docs/Test_Plan.pdf` to be executed with their *Actual Result* and *Test Result* columns filled in.
What happened: the test cases were automated and run by CI on every pull request, and each pull request lists its results. The Test Plan's *Actual Result* and *Test Result* columns are still empty. PR #74 names its tests `test_ut028_…` rather than by test-case ID alone, and PR #78 has no results table.
Corrective action: QA & Docs (Komal) fills both columns for every Sprint 1 and Sprint 2 test case from the CI runs before the development freeze on 30 Oct.
Affects: all Sprint 1 test cases; NFR-007 (every FR linked to a test case).
Status: closed on 3 Nov, later than planned and by a different member. The columns were not filled by 30 Oct. The Deliverable 3 split gave the job to Kshitij, who filled both columns for all 119 cases from one run of the full suite against the frozen build `2d74453`: 118 automated cases and IT_022, run by hand in a browser. All 119 pass. The result is `docs/Test_Plan.pdf` v4.0, which also applies the Deliverable 3 test-case corrections.

D-08 — Eleven more pull requests merged without a second member's approval

Baseline: NFR-006, and D-04's corrective action: from Sprint 2, every pull request, including the repository owner's, waits for an **Approve** review from another member.
What happened: the corrective action was not kept. After D-04 was recorded on 27 Oct, eleven of Kshitij's pull requests were merged to `main` with the repository owner's admin override. None had an **Approve** from another member when it was merged. The only review on each was GitHub Copilot's automated comment. There were still no direct commits to `main`, and CI passed on all eleven.

PR	Item	Open before merge	Review at merge	Merged
#82	Fix template tags split across lines	Minutes	Copilot comment only	27 Oct
#83	FR-024 total, percentage and Pass / Fail	13 h	Copilot comment only	28 Oct
#84	FR-021 resume an interrupted attempt	13 h	Copilot comment only	28 Oct
#86	Load-test review follow-ups (NFR-001, NFR-003)	1 h	Copilot comment only	28 Oct
#87	SQLite WAL fix; 200-user load test passes (NFR-001, NFR-003)	Minutes	Copilot comment only	28 Oct
#89	Sprint 2 data model (migration 0005, shared test data)	5 h	Copilot comment only; Kartik approved on 31 Oct, after the merge	30 Oct
#90	FR-009 version a question once it is in use	3 h	Copilot comment only	30 Oct
#91	FR-019 autosave, clear and mark for review	3 h	Copilot comment only	30 Oct
#92	NFR-002 evaluate 100 questions within 5 s	3 h	Copilot comment only	30 Oct
#93	NFR-004 service back within 60 s after a restart	3 h	Copilot comment only	30 Oct
#94	NFR-010 policy values configurable per exam	3 h	Copilot comment only	30 Oct

Why:

- **#82** fixed Django template tags that had been split across two lines. Django does not parse such tags, so they showed on the page as raw text. It was merged within minutes of being opened.
- **#86 and #87** were follow-ups to Komal's load-test pull request #85, which had been reviewed and approved. #87 is the fix that made the 200-user load test pass.
- **#83 and #84** were merged the same evening as #86 and #87, after 13 hours with no review.
- **#89 to #94** were merged on 30 Oct, the planned last day of Sprint 2, at the repository owner's direction. Both teammates' Sprint 2 branches build on #89: their tests use its shared test data (`portal.testkit.World`), and its migration adds the tables those features write to. The teammates' branches could not pass CI until #89 was on `main`.

Effect: NFR-006's acceptance test fails for these eleven pull requests. Every other pull request merged in Sprint 2 carries an **Approve** from a member other than its author: #81, #85, #88, #95, #97, #98, #99, #100, #101 and #102.
Note on #95, #99, #101 and #102 (merged 1 Nov): the repository owner resolved their merge conflicts with `main` on Kartik's and Komal's branches, at the team's request, and then approved them. NFR-006's test is met, because the reviewer is not the pull request's author. However, those conflict-resolution commits were reviewed only by the person who wrote them. Each pull request's review comment lists exactly what was changed.
Corrective action: this replaces D-04's. An admin merge stays possible when waiting would miss a deadline. Each one is recorded in this register on the day it happens, naming the pull request and why it could not wait for an **Approve**.
Recorded in: this register. The pull requests' review histories are the evidence.

D-09 — Two Sprint 2 items reassigned from Komal to Kshitij

Baseline: the Sprint Plan assigns NFR-007 (every FR linked to a test case, 1 point) and NFR-012 (lock an account after 5 failed sign-ins, 2 points) to Komal.
What happened: on 1 Nov, the last day of Sprint 2, Komal was still finishing FR-025/026 (PR #95) and FR-030 (PR #96), so Komal asked Kshitij to take these two items over. Kshitij completed both on the existing scaffold branches. Issues #58 and #62 were reassigned to Kshitij the same day.

Item	Issue	Branch	Delivered
NFR-007 traceability matrix: all 118 Test Plan cases mapped to their FRs	#58	`test/nfr-007-traceability`	1 Nov
NFR-012 sign-in lockout	#62	`feat/nfr-012-lockout`	1 Nov

Follow-ups done the same day:

- The Test Plan had no case for NFR-012. Version 3.1 adds IT_038 under UC-02, which makes 119 cases in all, and the lockout tests carry that ID.
- 21 Test Plan cases had no automated test. Five already had one under an unkeyed name (UT_028 to UT_031 and IT_030) and were renamed. Twelve were written: UT_011, UT_021, UT_032, UT_033, IT_015, IT_016, ST_004, ST_006, ST_007, ST_008, ST_009 and ST_011. NFR-007's check that every case is automated or marked manual now runs. It exempts only FR-030's four cases (UT_068, IT_036, IT_037, ST_013), which arrive with PR #96.

Effect: Komal's Sprint 2 points drop by 3 and Kshitij's rise by 3. The two pull requests still need an **Approve** from another member before they merge (NFR-006).
Recorded in: issues #58 and #62, and their pull requests.

D-10 — Two admin merges on 1 Nov, and one fix commit reviewed only by its author

Baseline: NFR-006, under D-08's corrective action: an admin merge is allowed only when waiting would miss a deadline, and is recorded here on the same day.
What happened:

PR	Item	Review at merge	Merged
#108	FR-030 follow-ups: NFR-007 gate without the FR-030 exemption; audit checks in UT_033 and ST_011	Copilot comment only; merged with the repository owner's admin override	1 Nov
#109	This entry (Deviations register v1.2)	Copilot comment only; merged with the repository owner's admin override	1 Nov

Why: 1 Nov is the last day of Sprint 2. #109 records #108's admin merge and itself, so no further register update is needed to cover it. #108 changes tests only: it removes the temporary FR-030 exemption from NFR-007's gate once PR #96 was on `main`, and adds back two audit-log assertions. CI passed, and the full suite (227 tests) passes on `main`.
Related, not an admin merge: PR #96 (Komal, FR-030) was approved by the repository owner and merged normally, so NFR-006's test is met. Its last commit, however, was the owner's own: it added the missing audit calls (question edits with the answer key, submission, evaluation, every changed examination field, results publication) and undid an editor's reformatting. That commit was reviewed only by its author, as with the conflict resolutions noted under D-08. The approval comment on #96 lists exactly what it changed.
Effect: with #96 and #108 on `main`, D-09's open point is closed: every Test Plan case is automated or marked manual, with no exemption.
Recorded in: this register, on the day of the merge.

D-11 — Deliverable 3 reports admin-merged on the due date

Baseline: NFR-006, under D-08's corrective action: an admin merge is allowed only when waiting would miss a deadline, and is recorded here on the same day.
What happened: pull request 110 was merged on 3 Nov with the repository owner's admin override, without an **Approve** from another member. It adds:

- Test Plan v4.0, with results for all 119 cases and the test-case corrections;
- the Unit Test Report, written by Komal and committed under Komal's name;

- the Code Coverage and Branch Coverage reports, with the `.coveragerc` that reproduces them;
- this register, v1.3.

Why: 3 Nov is the due date for Deliverable 3. The pull request changes documents and coverage configuration only, not application code. The build under test stays `2d74453`.
Recorded in: this register, on the day of the merge.

D-12 — System Validation Report: Kartik's v1.0 corrected by the owner's re-validation

Baseline: the Deliverable 3 split (briefs of 1 Nov) gives the System Validation Report to Kartik: the 13 system test cases walked through in a browser with screenshots, and the manual NFR checks. It was due to Kshitij by Mon 2 Nov, 20:00.
What happened: Kartik sent the report (v1.0) with its screenshots and his test-case review on 2 Nov, before the deadline. It ran the automated suite and walked five system cases by hand. On 3 Nov Kshitij reviewed it and re-ran all 13 system cases and the manual NFR checks through the real user interface in Chrome, Edge and Firefox. The re-run found one defect that v1.0 had missed (FR-028, Defect 1: false focus-loss events on question navigation; ST_013 fails). It also found that NFR-001 and NFR-003, reported as Pass, were never load-tested at 200 users. The report's §6 lists every correction.
Effect: the report is v1.1. Kartik remains its author (v1.0 is committed under his name in pull request 111), with Kshitij's review and second run added. Defect 1 stays open, because the build under test is frozen (D-13).
Process: pull request 111 went to `main` on 3 Nov with the repository owner's admin override, without an **Approve** from another member, under D-08's corrective action: the owner wanted the documents on `main` for the 4 Nov testing session. Kartik and Komal were asked to review it after the merge.
Recorded in: this register, on the day of the merge.

D-13 — Documents arranged to the course templates and moved to `docs/`; FR-028 defect carried forward; sprint videos added late

Baseline: the SRS (v3.0), SAD (v1.0) and Test Plan (v4.0) as submitted, held in `Documents/`.
What changed: the evaluation checklist and the final-demo FAQ, issued on 3 Nov, set requirements the documents did not yet meet:

- **folder:** every document and report moved from `Documents/` to `docs/`;
- **SRS v3.1:** arranged to IEEE 830 (Overall Description, definitions, references), the use case diagram embedded, and §3.3 Security Requirements added with three security objectives (SO-1 … SO-3) mapped to the ten security requirements. No requirement was added, removed or reworded;
- **SAD v2.0:** adds traceability (§3.3), the security architecture (§4) and a design section with three UML sequence diagrams, the API design and the error-handling protocols (§7);
- **Test Plan v4.1:** §1–§4 follow the course's test plan outline (scope, quality objective, roles, methodology, bug triage, suspension criteria, completeness, deliverables, tools and environment), and §5.1 Security Validation is added. The 119 cases and their results are unchanged.

Correction found while writing SAD v2.0: SAD v1.0 said a `close_expired_attempts` command run by cron would submit attempts nobody opens again. It was never built: every request on an attempt submits it first if it has expired, stamped at its true expiry. SAD v2.0 §5.1 and §6 now describe what was built and the limit it leaves.
Defect carried forward: Defect 1 of the System Validation Report (FR-028) is not fixed on the frozen build `2d74453`. Test Plan v4.1 therefore records ST_013 as **Fail** (118 of 119 Pass), where v4.0, which relied on the automated test alone, had it as Pass. The proposed fix in `focus.js` and its browser test are the first item for the optional Sprint 3.
Sprint videos: the 1-minute Sprint 1 and 2-minute Sprint 2 outcome videos were not in the repository at the end of either sprint. Both were recorded on 3 Nov from the frozen build and added under `docs/videos/`. They are screen recordings of the real application, driven by a script, with captions naming each requirement shown.
Code touched: none, apart from one path in the docstring of `portal/traceability.py`.

Recorded in: this register, on 3 Nov.

D-14 — Sprint 2 ran two days past its end, and the development freeze moved from 30 Oct to 1 Nov

Baseline: the Sprint Plan runs Sprint 2 from 26 to 30 Oct and freezes development at its end: "after that only documentation changes" (§4).
What happened: 23 of the 45 Sprint 2 points were merged after 30 Oct. Every item was opened as a pull request on 30 or 31 Oct:

Item	Points	Owner	Pull request	Merged
NFR-008 consent and erasure	2	Kartik	#98	31 Oct
FR-027 scorecard with per-question marks	2	Kartik	#100	31 Oct
FR-018 one question at a time, palette	3	Kartik	#97	1 Nov
FR-028 focus-loss events	2	Kartik	#99	1 Nov
NFR-005 three browsers, 360 px	2	Kartik	#101	1 Nov
NFR-009 WCAG 2.1 AA on the attempt flow	2	Kartik	#102	1 Nov
FR-025 / FR-026 withhold and publish results	4	Komal	#95	1 Nov
FR-030 append-only audit trail	3	Komal	#96, follow-ups #108	1 Nov
NFR-012 sign-in lockout	2	Kshitij (D-09)	#104	1 Nov
NFR-007 every FR linked to a test case	1	Kshitij (D-09)	#105	1 Nov

PR #106 (a flaky assertion in IT_032's test) also went in on 1 Nov. The last merge of 1 Nov, #109, gives commit `2d74453`, the build every Deliverable 3 report tests.
Why: the items were coded inside the sprint but conflicted with one another: FR-018, FR-028, NFR-005 and NFR-009 all change the attempt page, and FR-025/026 and FR-030 both change the results and audit code. Each branch merged `main` into itself and was re-tested before its own merge (D-05's corrective action), which serialised them.
Effect: Sprint 2 is shown as 26 Oct – 1 Nov in the README and in this register's header, and D-09 calls 1 Nov "the last day of Sprint 2". The freeze is the build `2d74453` of 1 Nov. After it, only documents changed until D-15. All 45 Sprint 2 points are Done on the board.
Not recorded until now: D-05 recorded Sprint 1's overrun when it happened; Sprint 2's was missed, and was found in the review before the final demo.
Recorded in: this register, on 11 Nov.

D-15 — Defect 1 (FR-028) fixed after the freeze

Baseline: the frozen build `2d74453` (D-14), on which Defect 1 of the System Validation Report was left open and ST_013 recorded as **Fail** (D-13). On 11 Nov the instructor allowed changes before the final demo, on condition that each one is logged here.
What changed: `attempts/static/attempts/focus.js`. The browser hides a page while it unloads it, so the script reported every move to another question, and the submission, as a focus loss. It now notes the page's `pagehide` event, which comes before the unload's `visibilitychange`, and reports nothing after it. `pageshow` clears the note if the browser restores the page from its back-forward cache. A tab switch, a minimised window or another application taking focus fire no `pagehide`, so each is still reported once, stamped by the server (UT_067).
Check added: `integrity/browser_check_fr028.py` drives Chrome and Firefox through 3 focus losses (2 window blurs, 1 hidden tab) and 4 moves through the palette, then counts the stored events. Before the fix: 7 in each browser, the same count as the Defect 1 report. After: 3 in each. The Django suite cannot run this check, because its test client runs no JavaScript, and CI does not install a browser, so it is run by hand.
Limit it leaves: closing the examination tab, or loading another site in it, also unloads the page, so neither is reported now. The attempt stays open on the server, and the Student resumes it on return (FR-020). FR-028 names tab switches; this is recorded so that the integrity page is not read as covering more than that.
Effect: Defect 1 is closed. ST_013 passes in the browser, and the Test Plan stands at 119 of 119 (v4.2). The System Validation Report is v1.2. The build for the final demo is the merge of this change, not `2d74453`.
Recorded in: this register, on 11 Nov.